Topic 5: The Water Cycle and Water Insecurity

Overview

Water plays a key role in supporting life on earth. The water cycle operates at a variety of spatial scales and also at short- and long-term timescales, from global to local. Physical processes control the circulation of water between the stores on land, in the oceans, in the cryosphere, and the atmosphere. Changes to the most important stores of water are a result of both physical and human processes.

Water insecurity is becoming a global issue with serious consequences and there is a range of different approaches to managing water supply.

Content

Enquiry question 1: What are the processes operating within the hydrological cycle from global to local scale?		
Key idea	**Detailed content**	
5.1 The global hydrological cycle is of enormous importance to life on earth	a.	The global hydrological cycle's operation as a closed system (processes, stores and flows) driven by solar energy and gravitational potential energy. (1)
	b.	The relative importance and size (percentage contribution) of the water stores (oceans, atmosphere, biosphere, cryosphere, groundwater and surface water) and annual fluxes between atmosphere, ocean and land.
	c.	The global water budget limits water available for human use and water stores have different residence times; some stores are non-renewable (fossil water or cryosphere losses).